Hugo C. Sámano-Sánchez

Curriculum Vitae

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Summary

Mostly interested in evolutionary biology, structural bioinformatics, and AI-driven approaches to decode protein-protein interactions and mechanisms of pathogenesis. Trained in Genomic Sciences, I soon got into Bioinformatics and Computational Biology. In 2013, I started a Master's degree at the University of Basel and completed my thesis at the National University of Singapore working with lipidomics of malaria. I defended my PhD thesis at the University of Heidelberg after working with Short Linear Motifs at the host-pathogen interface at EMBL Heidelberg.
In my recent independent research group at the Zhejiang University-University of Edinburgh Institute, I lead projects that bridge fundamental evolutionary questions with translational applications in disease. I am excited to collaborate with groups at the intersection of cell signaling, infection, and big data to advance both basic research and clinical outcomes.

Appointments

10/24 - current **Assistant Professor, tenure track**, Zhejiang University-University of Edinburgh Institute, Haining, China

01/26 - current **Laitong Residencial College mentor**, Zhejiang University, Haining, China

09/25 - current **ZJE Programme Co-ordination Group member**, Zhejiang University-University of Edinburgh Institute, Haining, China

05/22 - current **PhD Supervisor**, *College of Medicine and Veterinary Medicine*, University of Edinburgh, Scotland

09/21 - current **Honorary Lecturer**, *College of Medicine and Veterinary Medicine*, University of Edinburgh, Scotland

Education

10/15 - 12/19 **Ph.D. in Molecular Biology**, European Molecular Biology Laboratory / University of Heidelberg, Heidelberg, Germany
Magna cum laude.

- Advisor: Dr. Toby J. Gibson.
- Computational screening for Eukaryotic Short Linear Motifs mimicked by bacterial proteins. Five publications.
- Computational prediction of Linear Motifs at the *Plasmodium*-Host interface.

09/13 - 02/15 **Joint Master Programme in Infectious Diseases, Vaccinology and Drug Discovery**, University of Basel, Switzerland and the National University of Singapore, Singapore

- Advisors: Dr. Bruce M. Russell and Prof. Markus R. Wenk.
- Data analysis of Mass Spectrometry and Tandem Mass Spectrometry experiments to identify characteristic lipids of *Plasmodium vivax*-infected human reticulocytes. Thesis.
- Bioinformatics analyses to study receptor-ligand interactions during *Plasmodium vivax* reticulocyte infection. One publication.

08/07 - 06/11 **B.Sc. in Genomics Sciences**, National Autonomous University of Mexico, Cuernavaca, Mexico
With honours.

Professional Experience

10/24 - current **Assistant Professor**, Zhejiang University-University of Edinburgh Institute, Haining, China

05/20 - 09/24 **Lecturer**, *Biomedical Informatics bachelor's programme*, Zhejiang University-University of Edinburgh Institute, Haining, China

10/15 - 01/20 **PhD Student / Bridging Postdoctoral Fellow**, European Molecular Biology Laboratory, Heidelgerg, Germany

10/12 - 07/13 **Research Assistant**, National Institute of Public Health, Cuernavaca, Mexico

- Advisor: Dr. Jesus Martinez-Barnetche.
- Bioinformatics analysis of high-throughput sequencing data from human and murine lymphocyte antigen receptors challenged with inactivated viruses or bacterial pathogens, respectively. Three publications.

01/12 - 09/12 **Bioinformatics Consultant**, Winter Genomics (http://www.wintergenomics.com), Cuernavaca, Mexico

- Advisor: Dr. Enrique Morett.
- Genome assembly, annotation and structural variation analysis of two *Babesia* species under two growth conditions. Bioinformatics analysis service at the Winter Genomics start-up.

Scientific Publications

* Contributed equally.

- Wang, H., Zhang, J., Zhang, N., Ji Y., Hong, X, **Sámano-Sánchez, H.**, Xu, F., Han, Y., Xiong, X., Welburn, S.C., Schulze, H., Bachmann, T.T., Ge, Y., Scharf, D.H. Antifungal resistance landscapes in major fungal pathogens in Zhejiang, China. *Fungal Biology and Biotechnology*, Accepted, 2026. doi: 10.1186/s40694-026-00223-1
- Tang, Q., Lian, C., Han, X., Zhang, X., Wang, Z., Wang, B., Zhu, T., Lin, X., Wang, X., Xu, Y., Xiao, M., Wang, Z., Li, J., Chen, S., Wang, Y., Liu, Y., Li, S., Shen, Z., Lu, X., Han, X., Zhou, Y., Xiao, M., Ran, J., Cao, X., Xu, X., **Sámano-Sánchez, H.**, Rodríguez, A., Wang, L., Chen, S., Xu, Z., Zhang, S., Yang, N., Tang, Y., Liu, J. A Non-Canonical Role of SMAD4 in Regulating 3D Genome Architecture to Inhibit Lung Squamous Cell Carcinoma Development. *Advanced Science*, June, 2026, e75839. doi: 10.1002/advs.75839
- Sanogo, B., Echoru, I., Jones, C., Butala, C., Lin, D., Maiga, H., **Sámano-Sánchez, H.**, Sacko, M., Kassegne, K., Welburn, S.C. A reservoir at the gates: a scoping review of nonhuman mammalian hosts for human schistosomiasis in western Africa and the critical challenge for elimination. *Infectious Diseases of Poverty*, January, 2026. doi: 10.1186/s40249-025-01394-6
- Giménez, A., Del Giudice, M., López, P., Guaimas, F., **Sámano-Sánchez, H.**, Gibson, T., Chemes, L., Arregui, C., Ugalde, J., Czibener, C. Brucella NpeA is a secreted Type IV effector containing an N-WASP binding short linear motif that promotes niche formation. *mBio*, June, 2024. doi: 10.1128/mbio.00726-24
- **Sámano-Sánchez, H.**, Gibson, T.J., Chemes, L.B. Using Linear Motif Database Resources to Identify SH2 Domain Binders. *Methods Mol Biol*, 2023. doi: 10.1007/978-1-0716-3393-9_9
- Kumar, K., Michael, S., Alvarado-Valverde, J., Meszaros, B., **Sámano-Sánchez, H.**, Zeke, A., Dobson, L., Lazar, T., Ord, M., Nagpal, A., Farahi, N., Kaser, M., Kraleti, R., Davey, N.E., Pancsa, R., Chemes, L.B., Gibson, T.J. The eukaryotic linear motif resource: 2022 release. *Nucleic Acids Research*, 2022. doi: 10.1093/nar/gkab975
- Mezsaros, B., **Sámano-Sánchez, H.**, Alvarado-Valverde, J., Calyseva, J., Martinez-Perez, E., Alves, R., Shields, D.C., Kumar, M., Rippmann, F., Chemes, L.B., Gibson, T.J. Short linear motif candidates in the cell entry system used by SARS-CoV-2 and their potential therapeutic implications. *Science Signaling*, 2021. doi: 10.1126/scisignal.abd0334
- **Sámano-Sánchez, H.**, Gibson, T.J. Mimicry of Short Linear Motifs by Bacterial Pathogens: A Drugging Opportunity. *Trends in Biochemical Sciences*, 2020. doi: 10.1016/j.tibs.2020.03.003
- Kumar, M., Gouw, M., **Sámano-Sánchez, H.**, Pancsa, R., Glavina, J., Diakogianni, A., Alvarado-Valverde, J., Bukirova, D., Calyseva, J., Palopoli, N., Davey, N.E., Chemes, L.B., Gibson, T.J. ELM - the eukaryotic linear motif resource in 2020. *Nucleic Acids Research*, 2019. doi: 10.1093/nar/gkz1030
- Sampietro, D., **Sámano-Sánchez, H.**, Davey, N.E., Sharan, M., Meszaros, B., Gibson, T.J., Kumar, M. Conserved SQ and QS motifs in bacterial effectors suggest pathogen interplay with the ATM kinase family during infection. *biorXiv*, 2018. doi: 10.1101/364117
- Gouw, M., Michael, S., **Sámano-Sánchez, H.**, Kumar, M., Zeke, A., Lang, B., Bely B., Chemes, L.B., Davey, N.E., Deng, Z., Diella, F., Gurth, C.-M., Huber, A.-K., Kleinsorg, S., Schlegel, L.S., Palopoli, N., Roey, K.V., Altenberg, B., Remenyi, A., Dinkel, H., Gibson, J.T. The eukaryotic linear motif resource - 2018 update. *Nucleic Acids Research*. doi: 10.1093/nar/gkx1077
- Kosaisavee, V., Suwanarusk, R., Chua, A.C.Y., Kyle, D.E., Malleret, B., Zhang, R., Imwong, M., Imerbsin, R., Ubalee, R., **Sámano-Sánchez, H.**, Yeung, B.K.S., Ong, J.J.Y., Lombardini, E., Nosten, F., Tan, K.S.W., Bifani, P., Snounou, G., Rénia, L.,

Russell, B. Strict tropism for CD71+/CD234+ human reticulocytes limits the zoonotic potential of *Plasmodium cynomolgi*. *Blood*, 2017. doi: 10.1182/blood-2017-02-764787

- Gouw, M., **Sámano-Sánchez, H.**, Roey, K.V., Diella, F., Gibson, T.J., Dinkel, H. Exploring Short Linear Motifs Using the ELM Database and Tools. *Current Protocols in Bioinformatics*, 2017. doi: 10.1002/cpbi.26
- Godoy-Lozano, E.E., Tellez-Sosa, J., Sanchez-Gonzalez, G., **Sámano-Sánchez, H.**, Aguilar-Salgado, A., Salinas-Rodriguez, A., Cortina-Ceballos, B., Vivanco-Cid, H., Hernandez-Flores, K., Pfaff, J.M., Kahle, K.M., Doranz, B.J., Gomez-Barreto, R.E., Valdovinos-Torres, H., Lopez-Martinez, I., Rodriguez, M.H., Martínez-Barnetche, J. Lower IgG somatic hypermutation rates during acute dengue virus infection is compatible with a germinal center-independent B cell response. *Genome Medicine*, 2016. doi: 10.1186/s13073-016-0276-1
- Cortina-Ceballos, B., Godoy-Lozano, E.E., Tellez-Sosa, M., Ovilla-Munoz, M., **Sámano-Sánchez, H.**, Aguilar-Salgado, A., Gomez-Barreto, R.E., Valdovinos-Torres, H., Lopez-Martinez, I., Aparicio-Antonio, R., Rodriguez, M.H., Martínez-Barnetche, J. Longitudinal analysis of the peripheral B cell repertoire reveals unique effects of immunization with a new Influenza virus strain. *Genome Medicine*, 2015. doi: 10.1186/s13073-015-0239-y
- Cortina-Ceballos, B.*****, Godoy-Lozano, E.E.*****, **Sámano-Sánchez, H.***, Aguilar-Salgado, A., Velasco-Herrera, M.D.C., Vargas-Chavez, C., Velazquez-Ramirez, D., Romero, G., Moreno, J., Tellez-Sosa, J., Martínez-Barnetche, J. Reconstructing and mining the B cell repertoire with ImmunediveRsity. *mAbs*, 2015. doi: 10.1080/19420862.2015.1026502
- Rosenblueth, M., Sayavedra, L., **Sámano-Sánchez, H.**, Roth, A., Martínez-Romero, E. Evolutionary Relationships of Flavobacteria and Enterobacterial Endosymbionts with their Scale Insect Hosts (Hemiptera: Coccoidea). *Journal of Evolutionary Biology*, 2011. doi: 10.1111/j.1420-9101.2012.02611.x

Awards and Scholarships

- Zhejiang University Teaching Achievement Award (2025).
- Second prize in the Teaching Competition for young faculty of Zhejiang University-University of Edinburgh Institute (2022).

Talks and Posters

Invited talks

- Institute seminar at the Biomedical Research Institute, National Autonomous University of Mexico, CDMX, Mexico, 2026.
- Talk for the BioLinkX seminar series, Zhejiang University Student Bioinformatics Association, Hangzhou, China, 2023.
- Talk to the Parasitology Unit of the Center for Infectious Diseases, Heidelberg University Hospital, Heidelberg, Germany, 2019.
- Institute seminar at the Biotechnology Research Institute (IIB), National University of San Martin (UNSAM), Buenos Aires, Argentina, 2017.

Selected talks

- Cold Spring Harbor Asia Conference: Bacterial Infection and Host Defense, Suzhou, China, 2019.
- VIII Argentinian Bioinformatics and Computational Biology Congress, Posadas, Argentina, 2017.
- 5th International Conference of Research on *Plasmodium vivax* Malaria, Bali, Indonesia, 2015.

Poster presentations

- BioMalPar XV: Biology and Pathology of the Malaria Parasite, Heidelberg, Germany, 2019.
- EMBO Workshop on Molecular advances and parasite strategies in host infection, Les Embiez Island, France, 2018.
- EMBO Conference on Hijacking host signalling and epigenetic mimicry during infections, Paris, France, 2017.
- 2nd Interdisciplinary signaling workshop, Visegrad, Hungary, 2017.
- The modularity of signaling proteins and networks, Seefeld, Austria, 2016.

Teaching

Mentor

- PhD supervisor (two students), dual degree ZJU-UoE Integrative Biomedical Sciences programme, currently.
- PhD co-supervisor (three students), University of Edinburgh Biomedical Sciences programme, 2021-2026.
- PhD co-supervisor (five students), dual degree ZJU-UoE Integrative Biomedical Sciences programme, 2021-2026.
- Master thesis supervisor (one student), University of Edinburgh Biomedical Sciences programme, 2023.

Course Organizer

- Biomedical Informatics 3 (BMI3), Biomedical Informatics Programme, Zhejiang University-University of Edinburgh Institute, China, 2023 to 2025. This is a core course on algorithm design.

- Integrative Biomedical Sciences 3 (IBMS3), Biomedical Informatics and Biomedical Sciences Programmes, Zhejiang University-University of Edinburgh Institute, China, 2022. This is a core course on career development and experimental design.
- Bioinformatics Tools for Protein Structure, Disorder and Interaction Analysis, National University of San Martin, Buenos Aires, Argentina, 2017.

Lecturer

- Advanced Mathematics 1 (AM1), Introduction to Biomedical Informatics 1 (IBI1), Applied Data Sciences 2 (ADS2), Genomics and Proteomics 2 (GP2), Infection 3 (IN3), Computational Biology and Systems Biology 3 (CBSB3), Computational Modelling and Machine Learning (CMML3), and Integrative Biomedical Sciences 4 (IBMS4) courses of the Biomedical Informatics and Biomedical Sciences Programmes, Zhejiang University-University of Edinburgh Institute, China, 2020-2026.
- Biomedical Disorders 1 and 2 courses of the Integrative Biomedical Sciences PhD programme, Zhejiang University-University of Edinburgh Institute, China, 2021-2026.
- Biomedical Frontiers course of the Artificial Intelligence and Digital Health MSc. programme, Zhejiang University, China, 2023.
- Basic Discrete Mathematics. Zhejiang University-University of Illinois Urbana-Champaign (ZJU-UIUC) Institute, China, 2024.

Skills and Interests

Languages	Spanish (native), English (advanced), French (basic), Chinese (entry level)
Research interests	Host-Pathogen interactions, Evolution, Protein Modeling, Algorithm development